Component list

Перечень элементов (BOM) - PDF (Поверхностный монтаж, Вертикальные индикаторы) for Variant [SMD Variation - Vertical Tubes] of Project [Nixie Clock Avito x4.PrjPcb]

Project: **Nixie Clock Avito x4.PrjPcb**
Variant: **SMD Variation - Vertical Tubes**
Author: Dimasssius
Report Date: 04.06.2026

#	Designator	Comment	Description	Footprint	Quantity
1	B1	CR1025	Battery Clip Coin, SMD, CR1025, Cell	CR1025	1
2	C1	1uF	Capacitor, Ceramic (MLCC), SMD, 1uF, 250V, ±10%, X7R, Multilayer Ceramic Capacitors MLCC - SMD/SMT, ROHS	CAP_THD_ELEC_ALUM_D8.0MM_H12.5MM_SHORT_SM	1
3	C2, C3, C4, C5, C10, C11	0.1uF	Capacitor, Ceramic (MLCC), SMD, 100nF, 6.3V, ±10%, X7R, Multilayer Ceramic Capacitors MLCC - SMD/SMT, ROHS	CAP_SMD_MLCC_C0603_(M1608)_C0402A_(M1005A)	6
4	C8	100uF	Capacitor, Aluminum Electrolytic Capacitor, SMD, D5xL5.4mm, 100uF, 6.3V, ±20%, Aluminum Electrolytic Capacitors - SMD, ROHS	CAP_THD_ELEC_ALUM_D4.0MM_H7.0MM_THD/SMD_SI	1
5	C9	0.1uF	Capacitor, Ceramic (MLCC), SMD, 100nF, 6.3V, ±10%, X7R, Multilayer Ceramic Capacitors MLCC - SMD/SMT, ROHS	CAP_SMD_MLCC_C0603_(M1608)_C0402A_(M1005A)_TI	1
6	D1	ES2J	Diode, SMD, SMA(DO-214AC), 600V, 2A, 35ns, 1.7V@2A, Fast Recovery / High Efficiency Diodes, ROHS	DO-41/DO-214AC_VERTICAL_SMD/THD_SMD	1
7	D2	SS12	Diode Schottky, SMD, SMA(DO-214AC), 20V, 1A, 500mV@1A, Schottky Diodes, ROHS	DO-41/DO-214AC_VERTICAL_SMD/THD_SMD	1
8	D3, D4	LED0603	LED Indication - Discrete, 0603, 20mA, 225mcd, Colorless Transparent Lens, -20℃~+85℃, Positive Stick, 620nm~630nm, Red, 120°, 50mW, 2.3V, LED Indication - Discrete, ROHS	LED 0603_(1608)_0402A_(1005)	2
9	IC2	K155ID1	Decoder, DIP-16, 5V±5%, 25mA, 150ns, High Voltage Output, BCD to Decimal Decoders, RoHS	DIP16	1
10	IC3, IC4, IC5, IC6, IC7	TLP291	Photovoltaic Output Optoisolator, SOP-4-1.27mm, 80V, 3.75kV, 50mA, 300mV@2.4mA, 8mA, 1, 5V, 1.25V, DC Transistor, Photovoltaic Output Optoisolators, ROHS	SOIC127P700X230-4N	5
11	IC9	DS3231SN	Real Time Clock, SOIC-16-300mil, I2C, Real Time Clocks, ROHS	SOIC127P1032X265-16N	1
12	IC10	ATMEGA328P-AU	Microcontroller, TQFP-32(7x7), AVR, 20MHz, Microcontrollers (MCU/MPU/SOC), ROHS	QFP80P900X900X120-32N WITH SPI INTERFACE 2.54	1
13	J1	Female Header 1x10-Pin	Female Header, 1x10-Pin, 10P, P=2.54mm, Direct Insert, H:8.5mm, W:2.54mm, D:3.2mm, Single Row, Black Brass, Top Square, Hole Plugin, 3A, 20mΩ, 1000V, -40℃~+105℃, Female Headers, ROHS	HEADER_FEMALE_10PIN	1
14	J2	Female Header 1x8-Pin	Female Header, 1x8-Pin, 8P, P=2.54mm, Direct Insert, H:8.5mm, W:2.54mm, D:3.2mm, Single Row, Black Brass, Top Square, Hole Plugin, 3A, 20mΩ, 1000V, -40℃~+105℃, Female Headers, ROHS	HEADER_FEMALE_8PIN	1
15	J3	TYPE-C 16PIN 2MD(073)	USB Connector, 5A, USB 3.1, Horizontal Attachment, 16P, Female, -25℃~+85℃, Type-C, SMD, USB Connectors, ROHS	USB_C_DOUBLE_V/H_HORIZONTAL	1
16	L1	FNR6045S151MT	Fixed Inductor, SMD, 6x6mm, 770mA, 150uH, Magnetic Shielded Inductor, ±20%, 880mA, 754mΩ, Fixed Inductors ROHS	XR6*8-151K-L_FNR6045S151MT_SMD	1
17	LS1	FUET-4020	Buzzer, SMD, 4x4mm, 65dB, Passive (Externally Driven), Electromagnetic, 4kHz, Buzzers ROHS	FUET-4020	1
18	Q1	STD20NF20	MOSFET (N-Channel), DPAK, 200V, 18A, 30W, 125mΩ@10V, 1 N-Channel, Single FETs, MOSFETs, RoHS	TO-220_THD/SMD_SMD	1
19	Q2	MMBTA42	Transistor Bipolar (BJT), NPN, SOT-23, 300V, 225mW, 25@10mA, 10V, 500mA, NPN Bipolar (BJT), ROHS	SOT105P255X140-3N	1
20	R1, R3, R6, R7, R15	10kΩ	Resistor, Thick Film Resistor, SMD, 10kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_THD/SMD_VERTICAL_R0603_R0402_10K_SMD	5
21	R2	1kΩ	Resistor, Thick Film Resistor, SMD, 1kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_THD/SMD_VERTICAL_R0603_R0402_1K_SMD	1
22	R4	680kΩ	Resistor, Thick Film Resistor, SMD, 680kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_THD/SMD_VERTICAL_R0603_R0402_680K_SMD	1
23	R5	100Ω	Resistor, Thick Film Resistor, SMD, 100Ω, ±5%, Chip Resistor - Surface Mount, ROHS	RES_THD/SMD_VERTICAL_R0603_R0402_100OHM_SMI	1
24	R8	470Ω	Resistor, Thick Film Resistor, SMD, 470Ω, ±5%, Chip Resistor - Surface Mount, ROHS	RES_THD/SMD_VERTICAL_R0603_R0402_470OHM_SMI	1
25	R9	4.7kΩ	Resistor, Thick Film Resistor, SMD, 4.7kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_THD/SMD_VERTICAL_R0603_R0402_4.7K_SMD	1
26	R10, R11	4.7kΩ	Resistor, Thick Film Resistor, SMD, 4.7kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	2
27	R12	3kΩ	Resistor, Thick Film Resistor, SMD, 3kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_THD/SMD_VERTICAL_R0603_R0402_3K_SMD	1
28	R13	33kΩ	Resistor, Thick Film Resistor, SMD, 33kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_THD/SMD_VERTICAL_R0603_R0402_33K_SMD	1
29	R14	56kΩ	Resistor, Thick Film Resistor, SMD, 56kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_THD/SMD_VERTICAL_R0603_R0402_56K_SMD	1
30	R16	10kΩ	Resistor, Thick Film Resistor, SMD, 10kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	1
31	R17	200Ω	Resistor, Thick Film Resistor, SMD, 200Ω, ±5%, Chip Resistor - Surface Mount, ROHS	RES_THD/SMD_VERTICAL_R0603_R0402_200OHM_SMI	1
32	R18	5.1kΩ	Resistor, Thick Film Resistor, SMD, 5.1kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	1
33	R19	10Ω	Resistor, Thick Film Resistor, SMD, 10Ω, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	1
34	R20	100Ω	Resistor, Thick Film Resistor, SMD, 100Ω, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	1
35	R21, R22, R23, R24	1kΩ	Resistor, Thick Film Resistor, SMD, 1kΩ, ±5%, Chip Resistor - Surface Mount, ROHS	RES_SMD_R0603_(M1608)_R0402A_(M1005A)	4
36	S1, S4, S7, S10	TS-1009B	Switch Tactile, Through Hole, 4.5x4.5mm, Round Button, 50mA, Bend-in 4.5mm, SPST, 12V, Plugin, Tactile Switches, ROHS	TS-1009B	4
37	S13	BME280	Development Board, LGA-8(2.5x2.5), Temperature and Humidity Sensor, ROHS	BME280_DEVBOARD	1
38	U1	CH340C	USB Converter, SOP-16, 2Mbps Transceiver, USB 2.0, USB Converters, ROHS	SOIC127P600X180-16N	1
39	Y2	CSTCE16M0V53-R0	Ceramic Resonators, SMD, SMD3213-3P, 3.2x1.3mm,16MHz, 15pF, 40Ω, Ceramic Resonators, ROHS	XTL_3_3213	1
Notes					**60**